Model-completed bathymetry and a shoal-hazard field for the Churchill trade corridor, validated on soundings the model never saw

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Interactive atlas: girardemilio3-svg.github.io/churchill-corridor-atlas · Code and results: github.com/girardemilio3-svg/seabednet-validation

PLANNING PRIOR — NOT FOR NAVIGATION

Abstract. Canada, Manitoba and Saskatchewan have committed C$262.5M to reopen the Churchill trade corridor through Hudson Bay and Hudson Strait, water in which the Canadian Hydrographic Service (CHS) reports 15.8% of Arctic waters and 44.7% of key routes surveyed to modern standards. Under the 2,327 km Churchill–Atlantic route itself, 17% of the seabed carries a published sounding; the median survey year of those soundings is 1974. We train a heteroscedastic masked-completion network on the public CHS NONNA archive and score it on two sets of depths it never saw. (1) 109,044 research-cruise multibeam cells on the shelf (50–400 m) at locations where NONNA has no sounding: mean absolute error 4.9 m, against 13.4 m for the nearest published sounding and 5.5 m for the gravity-derived prior. (2) A temporal split using CHS's own dated Survey Index: a model trained only on soundings from before 2016 predicts the 6.46 million post-2016 soundings in the corridor to 13.3 m (18.8 m nearest sounding, 18.1 m gravity), with 74% of errors inside the model's 1σ. Because ships ground on the shallowest point and not the mean, we further train a hazard model on NONNA-10/100 pairs to predict the minimum depth within 500 m (error 2.7 m on held-out tiles versus 11.0 m if the archive depth is taken as the shoal) and hindcast 7 Transportation Safety Board groundings using only pre-incident soundings: the strike site falls in the top 10% of hazard among apparently safe water in 5 of 7 cases. We publish a calibrated uncertainty field, a σ-ranked survey plan (219 ship-days, C$40M), a

sealed forecast of 1,314 predicted depths (SHA-256 on record), and the full code and validation cells. Limitations are stated in §8.

1. Problem

Hydrographic adequacy in the Canadian Arctic is a documented gap: the CHS, reporting through the IHO in December 2025, put 15.8% of Canadian Arctic waters and 44.7% of key routes at adequate survey standard [1]; in 2016 its Hydrographer-in-Charge for the Arctic put full charting "more than a decade" away [2]. Groundings on uncharted or poorly charted shoals recur — Hanseatic (1996), Clipper Adventurer (2010), Akademik Ioffe (2018), Thamesborg (2025) — and the TSB reports in each case identify the survey state of the water as a contributing factor [3–6].

The federal–provincial Churchill Plus commitment (C$262.5M, February 2026) funds port and rail, not charting. The question this report addresses is narrow: given only the public archive, how much of the corridor's seabed can be inferred with stated, tested uncertainty, and where should a survey ship go first? A second question follows from the first and is, to our knowledge, unaddressed in the completion literature: a completion model predicts mean depth, but a keel meets the shallowest point. We treat that quantity as a separate target.

2. Data

Archive. CHS NONNA-100 (100 m) and NONNA-10 (10 m) non-navigational bathymetry, Open Government Licence – Canada, retrieved on 29 August 2026 through the CHS WCS endpoint. 437 national 100 m blocks (≈2,000×2,000 cells) and 3,601 10 m tiles; corridor subset of 94 blocks covering Hudson Bay, Hudson Strait, the Labrador approaches and Franklin Strait. NONNA is the *published* archive; CHS holds surveys not yet released to it. Throughout, "no published sounding" is the only claim made; "unsurveyed" is never inferred.

Physics prior. SRTM15+ V2.7 gravity-derived bathymetry (Scripps) [7], resampled to a 295 MB Canada raster; used as an input channel and as the always-reported baseline.

Independent multibeam. Gridded multibeam from the GMRT synthesis (Lamont-Doherty) [8], `topo-mask` layer, which returns measured cells only. In the corridor bounding box these derive from NCEI-archived research cruises (CCGS Amundsen 2003–2013, USCGC

Healy, R/V Knorr, R/V Neil Armstrong, R/V Maria S. Merian; 36 surveys). Independence from NONNA is by provenance: research-cruise data are not CHS holdings.

Survey dates. The CHS Survey Index (DFO EGIS), 7,078 polygons with survey start/end and CATZOC grade, 1832–2016 [9]. Rasterized onto every block, it dates each sounding to the latest indexed survey covering it; a sounding outside every polygon is, by construction, post-2016. Nationally, 46.8% of NONNA-100 soundings and 46.5% of NONNA-10 soundings are post-index; in the corridor, 19,471,411 of 34,281,587 (56.8%).

Groundings. Positions, dates and drafts from TSB reports M96H0016, M10H0006, M12H0012, M14C0219, M18C0225 and the open M25C0241 occurrence [3–6]; the Thamesborg position is from AIS and press reporting and is flagged as such.

3. Method

3.1 Completion model (mean depth and uncertainty)

A hierarchical ConvNeXt-style U-Net with self-attention at the two coarsest scales, FiLM-conditioned on a learned resolution embedding (10 m / 100 m). Inputs per 256×256 patch: visible depth, visibility mask, gravity prior. All depth channels are normalized by the gravity patch statistics (mean, standard deviation floored at 5 m), so normalization is continuous across windows and the prediction is a residual over physics. Two heads: μ and $\log \sigma^2$, trained with the heteroscedastic Gaussian negative log-likelihood on hidden cells plus an L1 term on visible cells. Masks mix harvested real NONNA gap patterns, random blocks and half-plane survey edges. Configurations: tiny (6.8M parameters) and small (34.8M; widths 96–768, depths 2/3/6/3). AdamW 2×10^{-4}, cosine schedule, batch 16, bf16; 4,000 steps. Inference: overlapping windows (stride 128) with Hann blending; windows require ≥400 visible cells; predicted land ($\mu > -4$ m) is dropped; inference is published only within 6 km of a sounding.

3.2 Hazard model (shallowest point)

Where NONNA-10 and NONNA-100 overlap, the 10 m grid gives the sub-cell extreme statistics the 100 m grid hides. For each 100 m cell we compute the shallowest 10 m sounding within a 500 m radius (a 101×101 maximum filter on negative depths, requiring ≥50% coverage of the disc). 3,601 tiles mosaicked 4×4 give 475 groups of 800×800 cells. Along the Churchill route the shallowest point sits a median 6 m above the 100 m mean. The same backbone is trained to predict this extreme and its σ from the masked 100 m field and gravity, everywhere — the extreme is unknown even where a 100 m sounding exists. The hazard for a draft *d* is

```
P(shoal < d) = 1 − Φ( (−d − μ_s) / σ_s )
```

Groups within 0.4° of any of the six grounding sites and the three geographic holdouts are excluded from training. Configurations: tiny (1,500 steps) and small (3,000 steps).

3.3 Validation design

Test 1 (independent). Cells with a GMRT measured depth, no NONNA sounding, and a model fill within 60 cells (6 km) of a sounding; baselines are the nearest published sounding (via distance transform) and the gravity prior; stratified by distance and depth. **Test 2 (temporal).** The model is retrained with every post-index sounding blanked from the corpus (100 m and 10 m); it is then given the pre-2016 soundings of each corridor block and scored on the post-2016 ones. **σ calibration.** An isotonic map from raw σ to the 68th and 95th percentiles of |error| is fitted on Test 1 shelf cells and applied everywhere. **Hindcast.** For each grounding, the hazard model receives only soundings dated before the incident (all soundings for Thamesborg, since NONNA carries no post-2016 dates); the strike cell's P(shoal < draft), allowing 500 m of position slop, is ranked among "apparently safe" cells — mean-model depth deeper than twice the draft — within 25 km. A rank at or above the 90th percentile is counted as a landing.

4. Results: completion

4.1 Test 1 — research-cruise multibeam the archive does not contain

DISTANCE TO NEAREST SOUNDING	N CELLS	SEABEDNET (M)	NEAREST SOUNDING (M)	GRAVITY PRIOR (M)
0–0.5 km	37,313	4.8	8.5	5.9
0.5–1 km	31,657	4.6	12.9	5.5
1–2 km	27,743	5.0	17.5	4.8
2–4 km	12,151	5.5	20.8	5.8

DISTANCE TO NEAREST SOUNDING	N CELLS	SEABEDNET (M)	NEAREST SOUNDING (M)	GRAVITY PRIOR (M)
trend + natural-neighbour residual			11.2	
trend + inverse-distance residual			8.7	
GEBCO reference (not independent, §4.5)			3.0	
all shelf 50–400 m	**109,044**	**4.9**	**13.4**	**5.5**

Table 1. Mean absolute error on independent multibeam cells, shelf water 50–400 m, by distance from the nearest published sounding. 386,685 cells over 17 blocks in total, of which 109,044 on the shelf.

DEPTH BAND	N CELLS	SEABEDNET (M)	GRAVITY PRIOR (M)
coastal < 50 m (excluded, see §4.4)	5,858	21.4	11.8
shelf 50–400 m	109,044	4.9	5.5
slope 400–1000 m	5,849	29.5	6.4
basin > 1000 m	265,934	10.9	6.3

Table 2. The same test by depth. Below 400 m the gravity prior is the better estimator; the published completion therefore defers to it there (300–400 m blended).

On the shelf the model reduces error by 64% relative to the archive's own nearest sounding and the advantage grows with distance (5.5 vs 20.8 m at 2–4 km). Against the gravity prior the margin on these cells is small (4.9 vs 5.5 m): research cruises transit deep channels where gravity inversion already works, and the glacial shelf where it fails is where NONNA is dense and therefore absent from this test. Test 2 addresses that.

4.2 Test 2 — pre-2016 model, post-2016 soundings

DISTANCE TO NEAREST PRE-2016 SOUNDING	N CELLS	SEABEDNET (M)	NEAREST SOUNDING (M)	GRAVITY (M)	BIAS (M)	|ERR| ≤ 1Σ
0–0.5 km	998,636	7.1	5.6	17.8	-0.3	65%
0.5–1 km	910,868	10.9	11.8	17.9	+0.6	72%

DISTANCE TO NEAREST PRE-2016 SOUNDING	N CELLS	SEABEDNET (M)	NEAREST SOUNDING (M)	GRAVITY (M)	BIAS (M)	|ERR| ≤ 1Σ
1–2 km	1,642,606	13.3	17.1	17.5	-1.1	75%
2–4 km	2,543,006	15.6	25.4	18.2	+1.1	77%
4–8 km	365,496	20.1	34.9	20.7	+3.2	75%
trend + natural-neighbour residual			16.8			
trend + inverse-distance residual			16.2			
GEBCO reference (not independent, §4.5)			6.2			
all	**6,460,612**	**13.3**	**18.8**	**18.1**	**+0.4**	**74%**

Table 3. Temporal holdout, corridor, 34.8M-parameter model. Positive bias = model shallower than the sounding. The 6.8M model scored 13.5 m overall on the same cells; model capacity is not the limiting factor.

DEPTH BAND	N CELLS	SEABEDNET (M)	NEAREST (M)	GRAVITY (M)	BIAS (M)
0–20 m	118,374	15.3	21.1	11.1	-9.8
20–50 m	311,616	17.9	21.6	13.8	-11.0
50–100 m	672,069	13.9	17.4	15.3	-3.9
100–200 m	1,741,821	8.4	12.2	9.7	+0.3
200–400 m	2,346,496	11.9	17.2	17.2	+2.0
400–9999 m	1,254,668	20.8	31.1	34.7	+3.4

Table 4. Temporal holdout by depth. In water shallower than 50 m the mean-depth model reads 10–11 m too deep — the direction that matters for a keel, and the motivation for §5.

4.3 Calibration and deferral

Raw σ ranked error correctly (Spearman-like correlation 0.30 on Test 1 shelf cells) but was under-confident: 48% of errors inside 1σ where 68% is expected, 74% inside 2σ. The isotonic map (mean factor 1.9×) restores 68% / 95% coverage; the temporal test, scored with the raw σ of a differently trained model, independently shows 74% inside 1σ. All uncertainty on the atlas is the calibrated one. Inferred cells deeper than 400 m carry the gravity prior (Table 2).

RAW Σ DECILE (M)	N	|ERR| P50 (M)	|ERR| P90 (M)	INSIDE 1Σ
0.3–2.1	38,697	5.0	14.1	13%
2.1–3.0	38,749	4.1	12.3	32%
3.0–3.8	38,720	4.2	12.4	42%
3.8–4.9	38,719	3.7	11.7	56%
4.9–5.9	38,894	4.2	11.3	60%
5.9–7.2	38,908	4.8	12.2	64%
7.2–10.3	38,716	5.3	15.5	69%
10.3–17.8	38,808	6.9	21.7	76%
17.8–29.7	38,751	9.1	29.2	85%
29.7–135.6	38,705	19.3	76.1	79%

Table 5. Raw-σ calibration curve on all Test 1 cells.

4.4 Excluded cells

5,858 Test 1 cells shallower than 50 m, at a median 131 m from a sounding and all at the shoreline of three blocks, disagree with every source: the model reads -20 m, the nearest CHS sounding -39 m, and gravity +9 m relative to GMRT. We attribute this to GMRT's coastal grid rather than measured swath and exclude the band from the headline pending inspection; it is reported here rather than removed.

4.5 Stronger baselines, GEBCO, and gravity leakage

Nearest-sounding is a floor. The classical gap fill is a gravity trend plus interpolated residuals (sounding minus SRTM15+ at the input soundings), interpolated by Delaunay natural-neighbour-style linear interpolation or by inverse-distance-squared over the 12 nearest residuals. On Test 1 these score 11.2 and 8.7 m against the model's 4.9 m; on Test 2, 16.8 and 16.2 m against 13.3 m (Tables 1 and 3, italic rows). GEBCO (NCEI global mosaic, GEBCO 2024/25, 15 arc-second) is reported as the chart-world reference and scores 3.0 m (Test 1) and 6.2 m (Test 2) on the 100 m grid; it is not an independent baseline because it ingests NONNA-

100 through IBCAO v5 (the Arctic Seabed 2030 compilation) and the same NCEI cruise archive; its lower error on both tests is the residual of resampling data it already contains, not a prediction, and where it has no source soundings it reduces to SRTM15+, the gravity row.

Gravity leakage. SRTM15+ V2.7 (released April 2025) inherits the cumulative NCEI multibeam archive of its point releases. Its error on the Test 1 cells is 5.5 m against 15.2 m on 282,193 ordinary NONNA-sounded shelf cells in the same blocks; the prior has almost certainly seen the cruise data. This favours the gravity baseline, which the model still edges on Test 1, and the model, which takes gravity as an input, inherits part of it. Test 2 is therefore the clean comparison: those NONNA shelf soundings are below SRTM15+'s resolution, and there gravity scores 18.1 m against the model's 13.3 m.

5. Results: hazard field and grounding hindcast

On held-out tile groups (around the grounding sites and the geographic holdouts), the 34.8M hazard model predicts the shallowest point within 500 m to 2.7 m mean absolute error where a 100 m sounding exists (11.0 m if that sounding is taken as the shoal; 12.3 m gravity) and 8.2 m where no sounding exists (11.0 m nearest field); 81% of errors inside 1σ, 96% inside 2σ. The 6.8M model: 4.0 / 9.4 m.

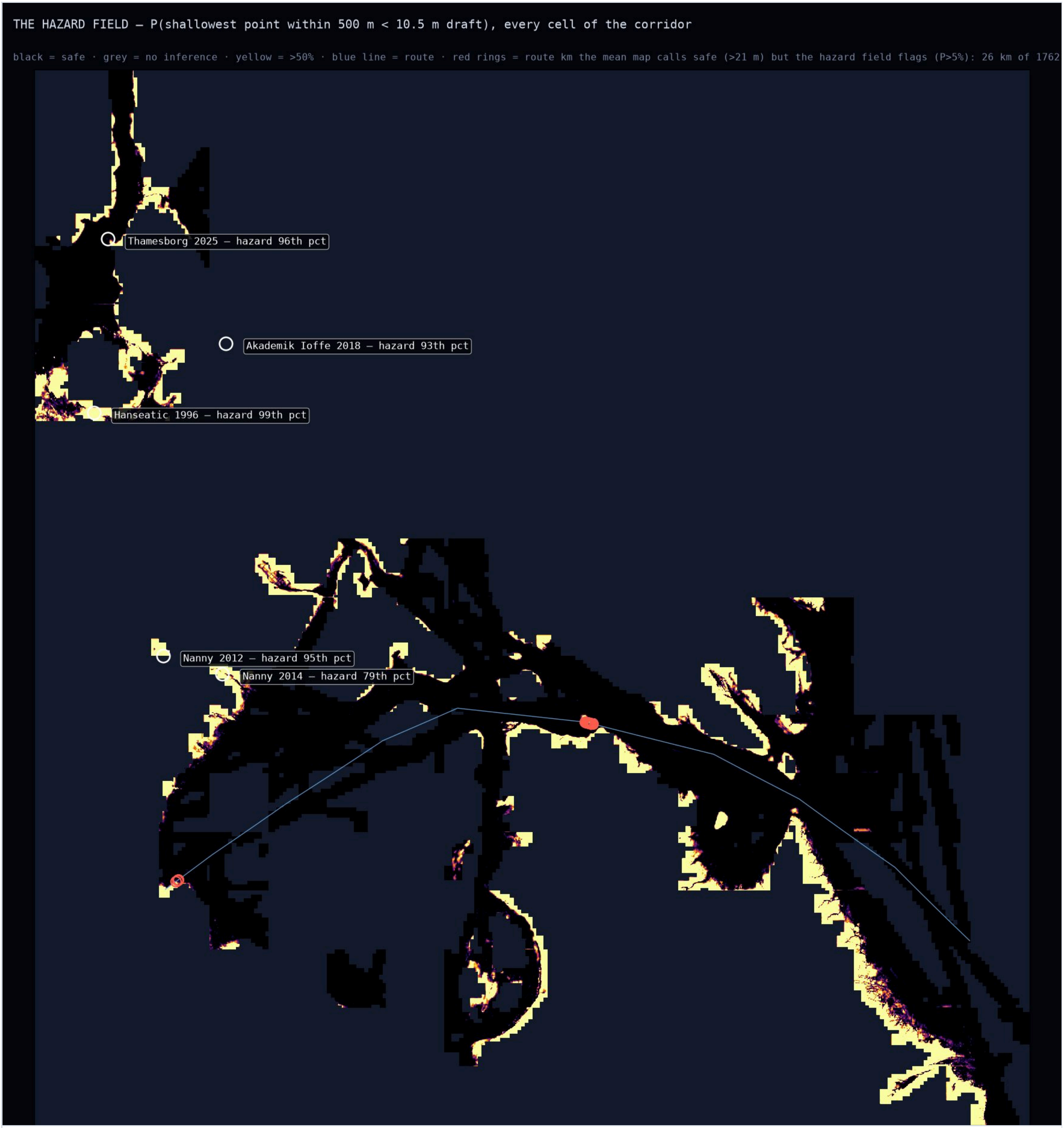

Figure 1. P(shallowest point within 500 m < 10.5 m) for every corridor cell. Blue: the Churchill route; red rings: the 26 route-km (of 1762 km deeper than 21 m on the mean map) where the hazard exceeds 5%; white rings: grounding sites with their hindcast percentile.

GROUNDING	PRE-INCIDENT SOUNDINGS IN BLOCK	KM TO NEAREST	P(SHOAL < DRAFT) AT SITE	HAZARD PERCENTILE	MEAN-Σ PERCENTILE	NEAREST-SOUNDING PERCENTILE
Thamesborg 2025-09-06 · TSB M25C0241	1,000,495	0.8	27%	**96**	99	90
Akademik Ioffe 2018-08-24 · TSB M18C0225	165,454	9.4	26%	**93**	77	90
Clipper Adventurer 2010-08-27 · TSB M10H0006	16,457	12.4	19%	**84**	11	22
Hanseatic 1996-08-29 · TSB M96H0016	148,672	0.3	94%	**99**	18	91
Nanny 2012 2012-10-25 · TSB M12H0012	12,501	6.3	95%	**95**	74	98
Nanny 2014 2014-10-14 · TSB M14C0219	36,222	0.0	62%	**79**	98	82
Mokami 2000 2000-10-31 · TSB M00N0098	208,248	0.0	82%	**95**	11	94

Table 6. Hindcast. Percentiles are among apparently-safe cells within 25 km; higher is more dangerous. Landings (≥90th): Thamesborg (96th), Akademik Ioffe (93rd), Hanseatic (99th), Nanny 2012 (95th), Mokami 2000 (95th). Akademik Ioffe and Clipper Adventurer lie 9–12 km from any pre-incident sounding, beyond the

6 km publication envelope; their scores are reported at a 12 km envelope. The two Nanny cases and Mokami 2000 (Bridges Passage, Labrador; buoy off station, datum mismatch) were navigation errors in known narrows and serve as controls; the hazard model's training did not exclude the Mokami tiles. The TSB counts 74 Arctic grounding/bottom-contact occurrences for 2000–2018, but publishes no occurrence-level positions; these 7 are every incident with a sourceable position. Four more are known by place name only (Nanny 2010, Mokami 2010, Dorsch 2012, Rosaire A. Desgagnés 2025) and are not scored.

The hazard percentile separates the four uncharted-shoal groundings from the baselines in a way neither the mean model's σ nor the nearest-sounding depth does: for Hanseatic the mean model's σ ranks the site at the 18th percentile (the area was densely sounded, so the mean model was confident) while the hazard field ranks it 99th. For Thamesborg the two agree (96th hazard, 99th σ), which is consistent with the court's finding that the ship was routed through CATZOC C water 1.7 km beyond the modern multibeam swath.

5.1 Blind inputs and sensitivity

Table 6 lets the model see every published sounding around a site at inference; NONNA carries no dates after 2016, so a post-incident survey could be among them. Table 6b removes every sounding within 10 km of each site from the input as well (the hazard model was trained with all tile groups within 0.4° of the sites excluded); a 25 km hole is also shown, at which the site lies outside the 12 km inference envelope and the model returns no estimate. The two denominator choices are then varied: comparison window (10/25/50 km) and the apparently-safe filter (1.5×, 2×, 3× draft on the mean map). Skill: by construction 10% of apparently-safe cells lie above the 90th percentile, so with the 10 km hole the 4-of-7 result has binomial $p = 0.0027$ (all six) and 3 of 4, $p = 0.0037$, on the four uncharted-shoal groundings; with all soundings visible, 5 of 7 ($p = 0.0002$). Precision cannot be bounded from six incidents; corridor-wide 18% of cells exceed $P = 5\%$ and 26 of 1762 apparently-safe route-km are flagged.

GROUNDING	ALL SOUNDINGS	10 KM HOLE	25 KM HOLE	10 KM HOLE · 10 KM WINDOW	10 KM HOLE · 50 KM WINDOW	10 KM HOLE · SAFE 1.5×	10 KM HOLE · SAFE 3×
Thamesborg	96	93	—	91	90	92	94
Akademik Ioffe	93	94	—	74	98	93	94
Clipper Adventurer	84	84	—	99	83	83	84

GROUNDING	ALL SOUNDINGS	10 KM HOLE	25 KM HOLE	10 KM HOLE · 10 KM WINDOW	10 KM HOLE · 50 KM WINDOW	10 KM HOLE · SAFE 1.5×	10 KM HOLE · SAFE 3×
Hanseatic	99	95	—	86	95	86	99
Nanny 2012	95	96	—	100	97	95	98
Nanny 2014	79	60	—	33	70	53	73
Mokami 2000	95	67	—	81	80	63	76

Table 6b. Hazard percentile of the strike site under input holes and denominator choices (window / safe-filter). 10 km hole with the 25 km window and 2× draft is the reference blind number; at 25 km the model has no input within its envelope and abstains.

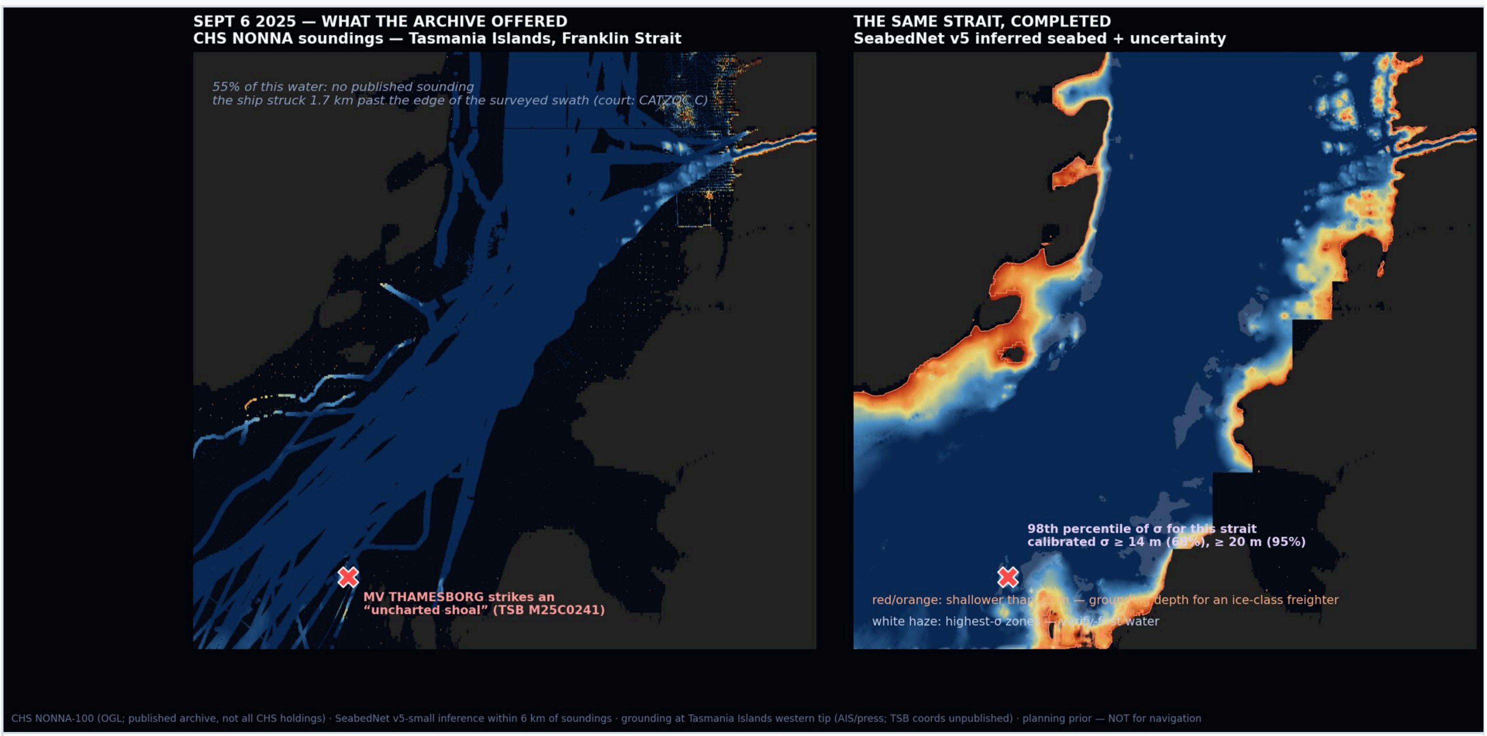

Figure 2. Franklin Strait, 6 September 2025. Left: published soundings (blue), the surveyed swath the court identified as the safer lane, and the strike position. Right: the model-completed depth with its uncertainty; the site sits at the 98th percentile of calibrated σ for the strait (≥ 14 m at 68%, ≥ 20 m at 95%). The nearest dated CHS survey is a 1960 single-beam line, CATZOC C.

6. The corridor product

Completion. 82,121 km² with published soundings → +351,305 km² inferred within 6 km

Route provenance. 1150 points at 2 km along the Churchill–Atlantic route: grade A (published of data (corridor); 429,682 → +1,049,018 km² nationally.

sounding) 17%, B (inferred, ≤1 km) 27%, C (1–6 km) 34%, D (no inference) 22%. Of the 198 sounded points, 103 (52%) date from before 1980; median survey year 1974.

Least-risk channel. Cost = distance × (shallow-water penalty + σ/8 + 13 × no-data); water under 20 m and land impassable. Result: 2,782 km, shallowest 22 m, mean σ 2.7 m, 0% over no-data water. It is 455 km longer than the straight route: about 20 h at 12 kn, ~21 t of fuel and a day of hire per voyage for a Handysize (~US$25k). It is a map of where the risk sits, not a sailing direction; the survey plan is what removes the detour.

Survey plan. Ten highest-σ boxes within 35 km of the route. Coverage: 40 km²/day, from a swath of ~3× depth (300–400 m at 100–130 m), 8 kn for 20 h (~300 line-km, ~100 km² raw), 20% overlap and a 50% weather-and-ice downtime factor; stated as an assumption. Day rate: the CCG polar-icebreaker charter of July 2026 (C$183,000/day; C$22M / ~120 days) [10] is the upper bound used in Table 7; the lower bound is the CHS Arctic hydrographic contract to Sedna ROV Services (Iqaluit), C$6.5M over two seasons, 9,000+ line-km in the first (DFO, March 2024): ~C$720 per line-km; at a 350 m swath with 20% overlap, ~C$2,600 per km², which prices the same area at about C$23M. CCGS Amundsen 2021: ~38,700 km² mapped over an ~80-day season (Copernicus ESSD 2024), i.e. several hundred km²/day in transit mapping of deep water; 40 km²/day is conservative for 100–130 m shelf water.

#	LAT	LON	INFERRED KM²	MEAN Σ (M)	PEAK Σ (M)	SHIP-DAYS	C$M
1	59.25°N	59.75°W	1,055	52.6	109.7	26.4	4.8
2	59.25°N	60.25°W	740	55.3	116.6	18.5	3.4
3	59.75°N	60.25°W	559	54.7	102.8	14.0	2.6
4	58.75°N	59.25°W	895	26.1	133.8	22.4	4.1
5	59.25°N	59.25°W	724	30.2	78.2	18.1	3.3
6	59.75°N	60.75°W	993	17.1	62.2	24.8	4.5
7	60.75°N	65.25°W	617	23.9	75.0	15.4	2.8
8	60.75°N	64.75°W	783	17.1	84.7	19.6	3.6
9	58.75°N	58.75°W	1,039	12.2	32.3	26.0	4.8

#	LAT	LON	INFERRED KM²	MEAN Σ (M)	PEAK Σ (M)	SHIP-DAYS	C$M
10	57.75°N	56.75°W	1,348	8.7	33.1	33.7	6.2

Table 7. σ-ranked survey plan: 219 ship-days, C$40.1M — one fundable season against "more than a decade" for full conventional coverage.

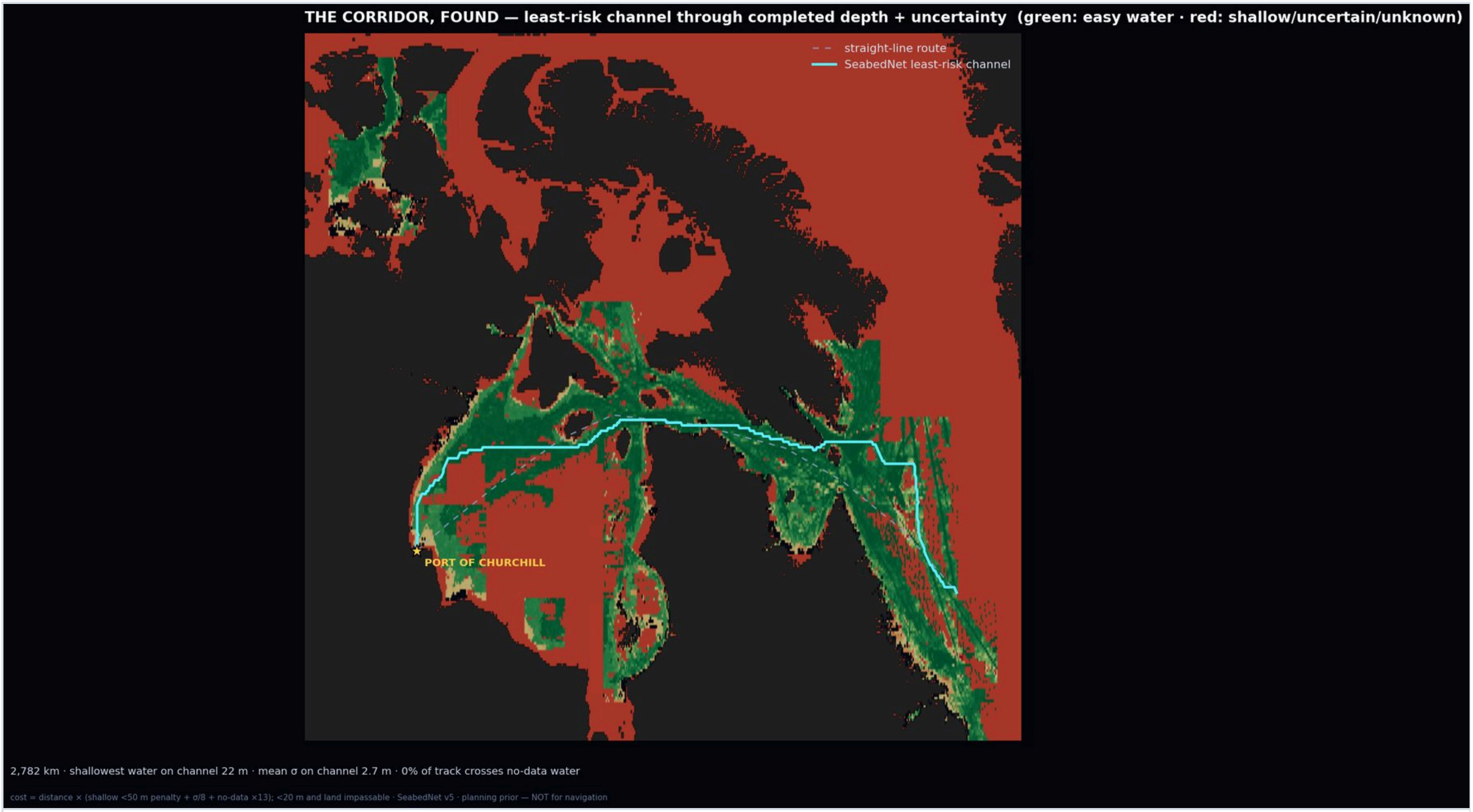

Figure 3. The least-risk channel (cyan) through the completed depth and uncertainty fields; dashed grey is the straight-line route. Green: easy water; red: shallow, uncertain or no-data.

7. A forecast that can fail

File `forecast_2026-09-02.csv` (1,314 cells: 314 on the found channel, the rest random inferred cells across the corridor) lists a predicted depth and calibrated 68% / 95% bands at cells that have no published sounding as of 29 August 2026. SHA-256 `4d3f5dc8a5c2b50f3f72fba95384581fc778d6d453a3eec46aa2dc383c66be45`, sealed 2026-09-02. Scoring rule: any later survey through these cells reports the mean absolute error and the fraction of soundings inside each band; we publish the score regardless of outcome. The depths were produced by the 34.8M-parameter v5-small completion model with the depth gate and calibrated σ of §4.3. The file is timestamped with OpenTimestamps (proof `forecast_2026-09-02.csv.ots`, three public calendars, Bitcoin-anchored) and its hash is in a GitHub commit of the validation repository, so the seal does not depend on our clock.

Grading. The forecast does not wait for a dedicated survey. On the first of every month the sealed cells' blocks are re-fetched from the CHS WCS; any sealed cell that has received a published sounding is scored (MAE, bias, band coverage, against nearest-sounding and gravity) and the result is published to the atlas with the per-cell record. The sealed file is hash-checked before every grade.

Benchmark. The temporal split of §3.3 is released as *NONNA-Temporal-Churchill v1*: index rasters for the 94 corridor blocks, the 6,686,037 target cells, the exact WCS request to rebuild the training side from public data, a scorer, and a leaderboard seeded with the baselines of §4.5 and SeabedNet. Anyone can try to beat it; submissions are added as declared.

8. Limitations

(i) NONNA is not the CHS holdings; a "no published sounding" cell may be surveyed. (ii) Test 1 under-represents the glacial shelf; Test 2 covers it but its labels are CHS's own later surveys, so datum and processing differences between survey eras are absorbed into the error. (iii) In water shallower than 50 m the mean-depth model is biased deep by 6–9 m (Table 4); the hazard model, not the mean model, is the appropriate product there. (iv) The hindcast has 7 cases, three of which are controls and two of which lie beyond the publication envelope; the TSB's own count is 74 occurrences for 2000–2018, so this is the public-coordinate subset, a benchmark and not a proof. (v) The Thamesborg position is not yet official. (vi) Survey-day rate and km²/day are assumptions from public contracts and stated arithmetic, not quotes. (ix) The gravity prior has very likely ingested the Test 1 cruises (§4.5). (vii) The physics prior itself ingests historical soundings, so the gravity baseline is not fully independent on surveyed water; on the cells of Test 1 it is. (viii) Nothing here is a chart. Every product carries "planning prior — not for navigation".

9. Reproducibility

All scripts, result JSON files and the per-cell validation arrays are public at github.com/girardemilio3-svg/seabednet-validation; the atlas page and this report are generated from those files by `build_atlas_v2.py`, `build_atlas_hazard.py` and `build_report.py`, so no number is transcribed by hand. Data are public (CHS NONNA under the Open Government Licence – Canada; GMRT; CHS Survey Index; SRTM15+). Training the 34.8M completion model to 4,000 steps takes 17 minutes on one RTX 5090 and about 12 hours on an NVIDIA GB10; the hazard model 13 minutes. Model weights are available on request.

10. Discoveries and sealed claims (3 September 2026)

National temporal test. The Test 2 protocol over all NONNA blocks: trained on pre-2016 soundings only, scored on 20,405,461 post-2016 soundings in 300 marine blocks across three oceans: model 12.6 m, nearest sounding 19.7 m, gravity 15.7 m; 74% of errors inside 1σ. On 539,352 cells in inland lakes and fjord heads the gravity anchor is invalid and the model fails with it (79 m vs 47 m nearest): the model is not applicable to inland water.

The Shoal List. The hazard model (§3.2, 34.8M) was run over all 437 blocks. Cells were kept where the completed depth is 21–150 m, P(shallowest point within 500 m < 10.5 m) ≥ 0.8, the predicted shallowest point lies between the surface and 21 m with relief under 100 m, the cell is within 6 km of a published sounding, no published sounding within ~500 m is shallower than 21 m, and the block's gravity prior agrees with its own soundings to 30 m (excluding inland lakes). Connected clusters of 3–300 cells: 626 nationally. The top 40 by probability, at most two per quarter-degree box, are sealed (`shoal_list_2026-09-03.csv`, SHA-256 a0bc8c3d2c174491…, OpenTimestamps). Each is a falsifiable claim scored by a single survey line. Regions: Strait of Belle Isle (7), Beaufort Sea (6), Hecate Strait (5), Chesterfield Inlet (5), Rae Strait / King William Is. (2), Gulf of St. Lawrence (2).

Strike-point predictions. For the four Arctic groundings known only by place name (Nanny 2010, Mokami 2010, Dorsch 2012, Rosaire A. Desgagnés 2025), the hazard model's most dangerous navigable cell in the reported area, clear of charted shore, was sealed before the TSB occurrence positions were requested (`strike_predictions_2026-09-03.csv`, SHA-256 a33994d6999ab30b…, OpenTimestamps); hit = within 2 km, graded verbatim on receipt.

Archive audits. Of 215,390,546 sounded cells nationally, 28.0% rest on surveys before 1980 and 8.1% before 1960; 46.8% are post-2016 (Figure 4). Community resupply lanes (15 km approach zones, 26 communities): the oldest medians are Naujaat 1955, Deception Bay 1960, Clyde River 1970; Milne Inlet, surveyed for the Baffinland ore trade, is 98% sounded with median 2017 and serves as the control. GEBCO lag: on 73,155 km² of CHS-sounded corridor water shallower than 200 m, the GEBCO-based global grid differs from the archive by more than 10 m on 14.4% of cells and is deep-biased by more than 10 m on 6.0% (4,523 km²); on the route, 15.7% of points differ by more than 10 m and 10.5% are deep-biased (worst 122 m). National σ-ranked survey plan: 39,569 km² in the 20 highest-uncertainty half-degree boxes, 989 ship-days, C$103–181M.

Negative results. ICESat-2 ATL24 laser bathymetry (SlideRule `atl24x`, ~3.6M photons over 12 corridor boxes): where the fitted datum offset is physical (±3 m) the laser agrees with NONNA soundings to 1–5 m, but only 124 photon cells without a published sounding survive the quality gates; the laser cannot score the model's fills in this turbid water. Era audit: post-2016 soundings within 200 m of a dated older sounding differ from it by a median of ~1 m in every decade back to 1900; the age of a sounding does not predict its error, and the risk of old charts lies in the unsounded water between soundings.

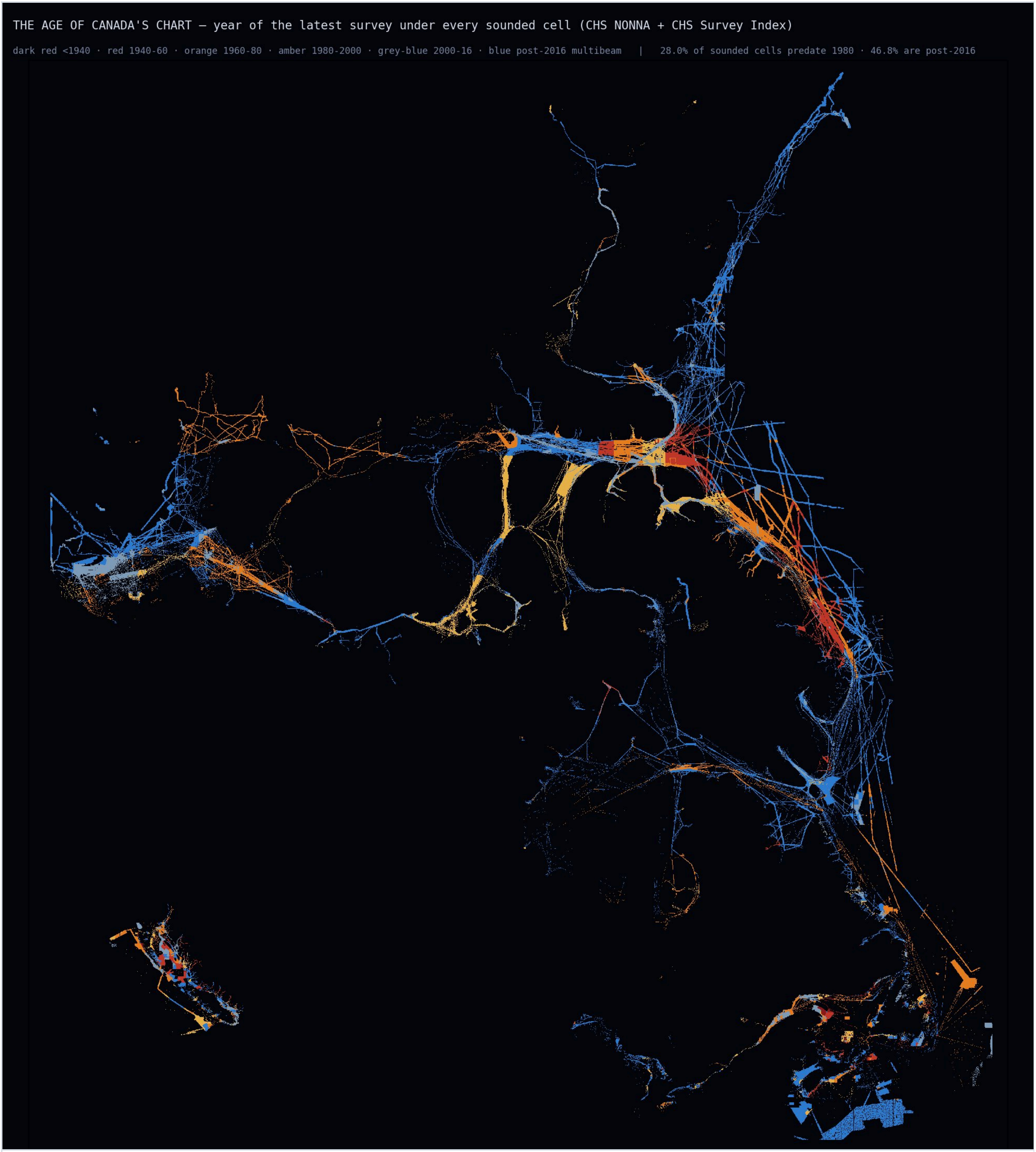

Figure 4. The age of Canada's chart: year of the latest survey under every sounded cell (CHS Survey Index; post-2016 = modern multibeam).

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10. Canadian Coast Guard polar-icebreaker charter, July 2026 (C$22M / ~120 days); Churchill Plus commitment, C$262.5M, February 2026; TC/MPO market-sounding study, February 2026.

Version 1.0, 2 September 2026. Generated from result files; see the repository for the exact commit.